AM 129: Big Data
HW5
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Problem 1: Murphy 12.5

(a) deriving PCA residual error

By the definition of the squared Euclidean norm,

$$\left\| x_i - \sum_{j=1}^{K} z_{ij} v_j \right\|^2 = x_i^T x_i - 2\sum_{j=1}^{K} z_{ij} x_i^T v_j + \sum_{j=1}^{K}\sum_{l=1}^{K} z_{ij} z_{il} v_j^T v_l .$$

Since $z_{ij} = x_i^T v_j$ and $v_j^T v_l = \delta_{jl}$,

$$= x_i^T x_i - 2\sum_{j=1}^{K} z_{ij}^2 + \sum_{j=1}^{K} z_{ij}^2$$

$$= x_i^T x_i - \sum_{j=1}^{K} z_{ij}^2$$

$$= x_i^T x_i - \sum_{j=1}^{K} v_j^T x_i x_i^T v_j$$

Hence,

$$\left\| x_i - \sum_{j=1}^{K} z_{ij} v_j \right\|^2 = x_i^T x_i - \sum_{j=1}^{K} v_j^T x_i x_i^T v_j . \quad \square$$

(b) express average reconstruction error using eigenvalues.

We want to prove $J_K = \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{K} \lambda_j .$

Proof. The average squared reconstruction error is

$$J_K = \frac{1}{n}\sum_{i=1}^{n} \left\| x_i - \sum_{j=1}^{K} z_{ij} v_j \right\|^2 .$$

Substituting the result from part (a),

$$J_K = \frac{1}{n}\sum_{i=1}^{n} \left(x_i^T x_i - \sum_{j=1}^{K} v_j^T x_i x_i^T v_j \right)$$

$$= \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{K} v_j^T \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T \right) v_j$$

For mean-centered observations, the covariance matrix is

$$\Sigma = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^T$$

Thus, $J_k = \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{k} v_j^T \Sigma v_j$.

Since v_j is an eigenvector of Σ, $\Sigma v_j = \lambda_j v_j$.

Therefore,

$v_j^T \Sigma v_j = \lambda_j v_j^T v_j = \lambda_j$, because $v_j^T v_j = 1$.

Substituting,

$$J_k = \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{k} \lambda_j. \quad \square$$

(c) residual error = sum discarded eigenvalues.

We want to prove $J_k = \sum_{j=k+1}^{d} \lambda_j$.

Proof. From part (b), $J_k = \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{k} \lambda_j$. When $k = d$, all principal components are retained, so the reconstruction error is zero:

$J_d = 0$. Therefore, $0 = \frac{1}{n}\sum_{i=1}^{n} x_i^T x_i - \sum_{j=1}^{d} \lambda_j$,

which implies $\frac{1}{n}\sum_{i=1}^{n} x_i^T x_i = \sum_{j=1}^{d} \lambda_j$.

Substituting into J_k:

$$J_k = \sum_{j=1}^{d} \lambda_j - \sum_{j=1}^{k} \lambda_j$$

$$= \left(\sum_{j=1}^{k} \lambda_j + \sum_{j=k+1}^{d} \lambda_j\right) - \sum_{j=1}^{k} \lambda_j$$

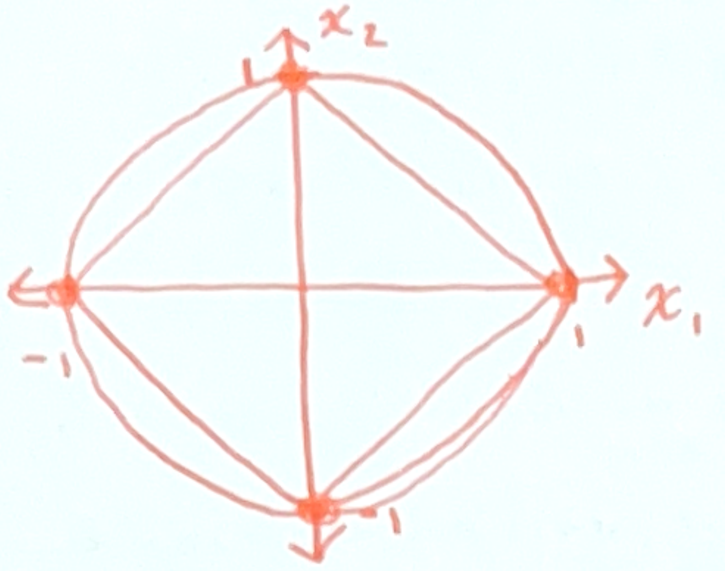
$$= \sum_{j=k+1}^{d} \lambda_j.$$

Hence, $J_k = \sum_{j=k+1}^{d} \lambda_j$. $\square$

Problem 2: ℓ_1-Regularization

(a) draw ℓ_1 and ℓ_2 norm balls for $K=1$

- For $x = (x_1, x_2)^T \in \mathbb{R}$, the ℓ_1 norm ball is $B_1 = \{x : |x_1| + |x_2| \leq 1\}$
- It's boundary is a diamond w/ vertices $(1,0)$, $(0,1)$, $(-1,0)$, $(0,-1)$.
- The Euclidean norm ball is $A_1 = \{x : x_1^2 + x_2^2 \leq 1\}$ which is the unit disk centered at the origin.



(b) connection constrained optimization & regularization

Consider the constrained optimization problem $\min_x f(x)$ subject to $\|x\|_p \leq K$.

We want to connect it to

$$\min_x [f(x) + \lambda \|x\|_p].$$

Proof. Rewrite the constraint as $g(x) = \|x\|_p - K \leq 0$.

Introduce a Lagrange multiplier $\lambda \geq 0$. The Lagrangian is

$$\mathcal{L}(x, \lambda) = f(x) + \lambda(\|x\|_p - K).$$

Expanding, $\mathcal{L}(x, \lambda) = f(x) + \lambda \|x\|_p - \lambda K$.

For a fixed λ, the term $-\lambda K$ is constant w/ respect to x. Thus,

$$\arg\min_x \mathcal{L}(x, \lambda) = \arg\min_x [f(x) + \lambda \|x\|_p].$$

Therefore, choosing the appropriate optimal Lagrange multiplier gives the regularized formulation

$$\min_x [f(x) + \lambda \|x\|_p].$$

Under convexity and suitable constraint qualifications, strong duality and the KKT conditions establish this correspondence for the constrained optimum. Different values of K correspond to appropriate choices of λ, although the relationship need not be one-to-one. □

(c) why l_1-regularization produces sparser solutions

The l_1 norm ball has a diamond shaped boundary with sharp corners on the coordinate axis, whereas the l_2 norm ball has a smooth circular boundary.

In constrained optimization, the optimum often occurs where a level curve of the object function first touches the feasible region. For l_1-regularization, this point is more likely to occur at a corner of the diamond, where one or more of coordinates are exactly zero. In contrast, the smooth boundary of the l_2 norm ball generally leads to solutions with nonzero coordinates that are merely reduced in magnitude.

Therefore, for suitably strong regularization, l_1-regularization promotes sparse solutions, whereas l_2-regularization generally shrinks coefficients toward zero without setting them exactly to zero.

Extra Credit: Lasso

(a) show that a laplace prior corresponds to l_1-regularization

Suppose a model has parameter vector $\theta = (\theta_1, \ldots, \theta_d)^T$. Assume each parameter has an independent zero-mean laplace prior w/ common scale $b > 0$: $p(\theta_j) = \frac{1}{2b} \exp\left(-\frac{|\theta_j|}{b}\right)$.

The joint prior is: $p(\theta) = \prod_{j=1}^{d} p(\theta_j)$

$$= \left(\frac{1}{2b}\right)^d \exp\left(-\frac{1}{b}\sum_{j=1}^{d}|\theta_j|\right)$$

$$= \left(\frac{1}{2b}\right)^d \exp\left(-\frac{1}{b}\|\theta\|_1\right).$$

By Bayes' Theorem, $p(\theta|D) = \dfrac{p(D|\theta)\,p(\theta)}{p(D)}$.

The maximum a posteriori estimate is $\theta_{MAP} = \operatorname{argmax}_\theta p(\theta|D)$.

Since $p(D)$ does not depend on θ, $\theta_{MAP} = \operatorname{argmax}_\theta p(D|\theta)\,p(\theta)$.

Taking the negative logarithm turns maximization into minimization:

$$\theta_{MAP} = \arg\min_{\theta} \left[-\log p(D|\theta) - \log p(\theta) \right].$$

Using the joint Laplace prior, $-\log p(\theta) = d\log(2b) + \frac{1}{b}\|\theta\|_1$.

Dropping the constant $d\log(2b)$, $\theta_{MAP} = \arg\min_{\theta} \left[-\log p(D|\theta) + \frac{1}{b}\|\theta\|_1 \right]$.

Setting $\lambda = 1/b$, we obtain, $\theta_{MAP} = \arg\min_{\theta} \left[-\log p(D|\theta) + \lambda\|\theta\|_1 \right]$

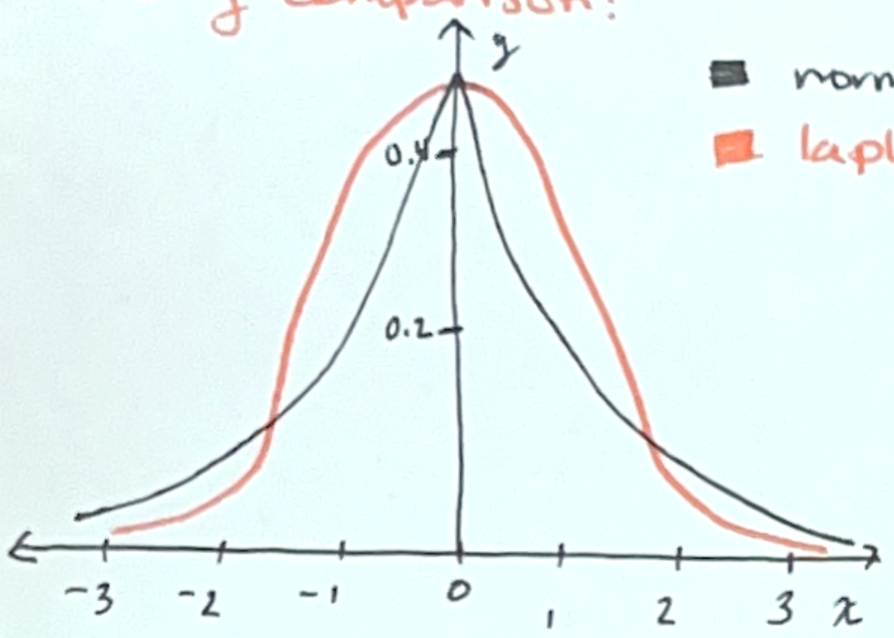
This is precisely the l_1-regularized maximum-likelihood objective.

Therefore, an independent zero-mean Laplace prior on the weights is equivalent to l_1-regularization in MAP estimation. □

(b) compare Laplace and standard normal densities

The two densities are $Lap(x|0,1) = \frac{1}{2}e^{-|x|}$ and $N(x|0,1) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

Density Comparison:



The Laplace density has a sharper peak at zero and heavier tails than the standard normal density.

(c) The Laplace distribution has a sharpr peak at zero and heavier tails than the Gaussian distribution. Consequently, a Laplace prior encourages many parameter estimates to be exactly zero while allowing some coefficients to remain relatively large.

The negative log of the Laplace prior produces an l_1 penalty, $-\log p(\theta) \propto \|\theta\|_1$, whereas a zero-mean Gaussian prior produces an l_2 penalty, $-\log p(\theta) \propto \|\theta\|_2^2$.

The l_1 penalty is not differentiable at zero, which permits the optimum to occur exactly at zero for a range of likelihood gradients. The smooth l_2 penalty generally shrinks coefficients continuously rather than forcing them to zero.

Thus, Laplace prior $\iff$ l_1 regularization $\Rightarrow$ sparsity.

In comparison, Gaussian prior $\iff$ l_2 regularization $\Rightarrow$ coefficient shrinkage

□.